Unknown Is Not Risky: Creative Decision Making with Calibrated Decision Models

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`cdm` v0.3.0 — exact decision theory on calibrated beliefs

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Abstract

Decision models select; they do not generate. We present `cdm`, a layered system that keeps generation (people or a generative model write the options) separate from selection (typed, calibrated judgments plus decision rules you control in code). The *belief layer* debiases, calibrates and pools judge outputs in Aitchison geometry; the *value layer* computes expected utility, CVaR, a fair-tie P(best), expected improvement and EVPI on finite lotteries; the *policy layer* gates each option as clear, verify or reject and then chooses to commit, reserve an innovation slot, spread a portfolio, or escalate. A fourth, *mechanism layer* borrows the description of a design from engineering, its functions and the working principle chosen for each, so that novelty is measured on how an option works rather than on how it is worded, and beliefs about whether a principle works are shared by every option that uses it.

The central design principle is that *uncertainty is not, by itself, a reason to reject*: an uncertain option is sent to a bounded check with a fallback to the status quo instead of being vetoed. We make this precise. For a single decision, the value of a check depends on the belief about the violation probability only through its mean, so ignorance and a known coin flip warrant the same action; the difference matters for learning across decisions. We model imperfect checks (sensitivity, specificity, cost, undetected harm), so residual risk is computed rather than assumed away, and give a maximin rule over every dependence between checks. Across decisions the shape of the belief does matter: one test of a principle with Beta(a, b) belief is worth at most $\bar{\theta}(1-\bar{\theta})/(n+1)$ per unit at stake in a later decision, so ignorance teaches 34 times more than a known coin flip. Every property is derived with its hypotheses; doing so corrected two over-claimed properties and a tie-handling bug that the tests had missed.

The evidence is kept apart by kind. (i) *Implementation*: 110 property-based tests and oracles at 99% branch coverage, and agreement with Monte Carlo. (ii) *A synthetic benchmark* with τ chosen on held-out problems: checking uncertain options beats vetoing them (regret 0.045 vs. 0.091, paired gain $[0.041, 0.050]$) at about 1% undetected harm, against 17–23% for acting on them unchecked. The gain holds under worse checks and a confidently wrong judge, but it comes from checking, not from the CVaR reliability gate, which the best policy leaves unbound. (iii) *A semi-synthetic benchmark with 237 real Jev calls* and ground truth by construction: Jev separates explicit harm from explicit safety almost perfectly (AUROC 0.9998), is calibrated on average on ambiguous options but varies with phrasing, and is more overconfident on value than its audit prior assumes. Over 48 problems `cdm` has the lowest regret and harm (gain over veto 0.051, bootstrap interval $[0.009, 0.102]$), with a gain that depends on an assumed status-quo value. None of this establishes better decisions on real outcomes, which we specify as the main open problem. (iv) *Creative engineering, live and simulated.* On 56 options built from known and new working principles, each written plainly and with hype, Jev names the principle in 99% of cases; hype raises its ordinal novelty by 0.058 $[0.038, 0.080]$ but the mechanism novelty

by 0.005 $[-0.006, 0.017]$. In a simulated programme of related decisions, learning per principle lowers regret in every setting tested and cuts late commitments to unreliable principles from 26% to 7%; the innovation slot finds more good principles but pays for itself in regret only when its pilots are small and the status quo is ordinary.

Keywords: calibrated decision models, decision theory, Aitchison geometry, CVaR, value of information, verification, position bias, morphological analysis, working principles, design novelty

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1 Introduction

A creative decision is a choice among options none of which is provably best. The interesting options are *novel*: there is little precedent, so little historical frequency to learn from, and the decision must rest on judgment. Much current tooling asks one model to propose an option, justify it and approve it. That conflates divergence (inventing options) with convergence (choosing one). A model that writes a persuasive rationale is not thereby calibrated: judgment models are often overconfident, sensitive to presentation order, and unable to say "I do not know" in a way a program can act on.

We separate the jobs. People or a generative model diverge; a pipeline of typed judgments converges through decision rules stated as math and checked by tests:

$$\text{options} \times \text{judges} \xrightarrow{\text{belief}} \Delta^{\circ}(\text{levels})^{D} \xrightarrow{\text{value}} \text{Lotteries}(\mathbb{R}) \xrightarrow{\text{policy}} (\text{commit}, \text{innovate}, \text{portfolio}, \text{escalate}).$$

The principle, stated carefully. The slogan in our title is shorthand for a narrower claim: *uncertainty is not, by itself, a reason for automatic rejection.* If an option can be checked cheaply, with a working fallback, then an unfamiliar option should be checked rather than vetoed, so the system can be conservative about irreversible commitment without being conservative about exploration. Two caveats are central and are developed in §4. First, a point estimate $q = 0.5$ cannot tell "we do not know" from "we know the harm rate is one half"; only the former is a reason to verify. Second, the downside is bounded only as well as the check works: a check that misses violations leaves residual risk, which we now model explicitly.

Contributions.

1. An architecture that makes selection a separate, inspectable process: judgments enter as calibrated distributions, and the decision rule is ordinary code with stated properties (§3).
2. A gate-and-fallback policy (clear / verify / reject, commit / innovate / portfolio / escalate) with properties P1–P8 derived under stated hypotheses, an explicit model of imperfect verification, the value of a one-shot check, and a maximin rule over dependence between checks (§4, Appendix A).
3. A mechanism layer from creative engineering: options classified into functions and working principles, a wording-invariant novelty measure, a prototype gate for unproven principles, beliefs per principle shared across options, and the value of one test for later decisions (Proposition 3) (§5).
4. An implementation whose claims are mapped to property-based tests; a synthetic benchmark with held-out tuning, ablation and robustness checks; a benchmark that runs the policy on real Jev judgments against known ground truth; a live test of whether Jev's novelty follows mechanism or wording; and a simulated programme of related decisions that evaluates learning and the innovation slot (§6).

The mathematical components are established; the contribution is their integration into one decision policy, and a clear account of what that policy does and does not guarantee.

2 Related work

Calibration. Temperature and Platt scaling correct systematic over- or under-confidence [14, 9]; they do not say how to act, and calibration measured in aggregate can hide miscalibration in sub-populations. *Combining judges.* Linear and logarithmic opinion pools [8] combine distributions;

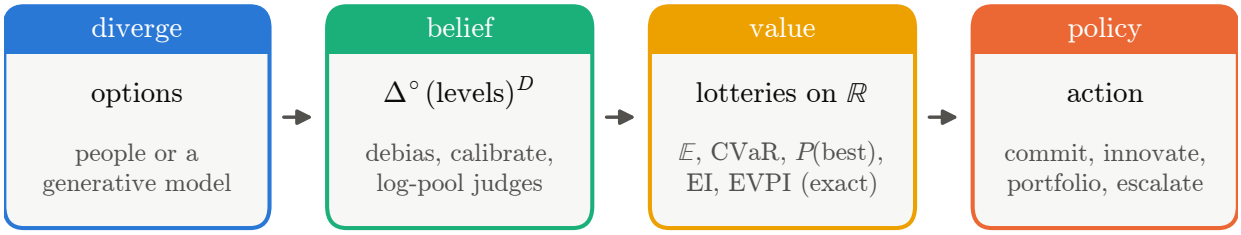


Figure 1: Generation is kept apart from selection. The belief layer turns judge outputs into calibrated distributions over levels; the value layer turns them into finite lotteries; the policy layer gates and selects. An option that fails verification falls back to the status quo B.

the log pool is externally Bayesian, but neither pool corrects for correlated errors. *Compositional data.* Aitchison geometry makes the open simplex a vector space [1], which turns calibration, pooling and Bayes updates into linear operations. *Presentation bias.* LLM judges are sensitive to option order [19]; permuting and averaging is the standard mitigation, and we state when it is exact. *Risk.* Coherent risk measures [4] and CVaR [15] quantify downside for a given distribution. *Exploration and information.* Expected improvement [12] trades exploration against exploitation; the value of information [11] prices further inquiry; conservative bandits [18] explore while keeping performance close to a baseline, which is close in spirit to our reliability gate. Online weights use Hedge [7]. *Creative engineering.* Design methodology describes a solution by its functions and the working principle chosen for each, and explores combinations in a morphological chart [20, 13]; TRIZ catalogues principles that resolve recurring contradictions [2], and C–K theory treats design as expanding concepts with new knowledge [10]. Ideation metrics score novelty by the rarity of each principle among known solutions [17], or by the level of abstraction at which an idea departs from its precedents [16]. Boden separates combinational from exploratory and transformational creativity [5]; Arthur describes technologies as combinations of mechanisms that harness phenomena [3]; and recombining familiar components is a main source of invention whose outcomes are more variable [6]. We use these descriptions as the input to decision theory: novelty and feasibility become properties of principles, which a calibrated judge classifies and a test resolves.

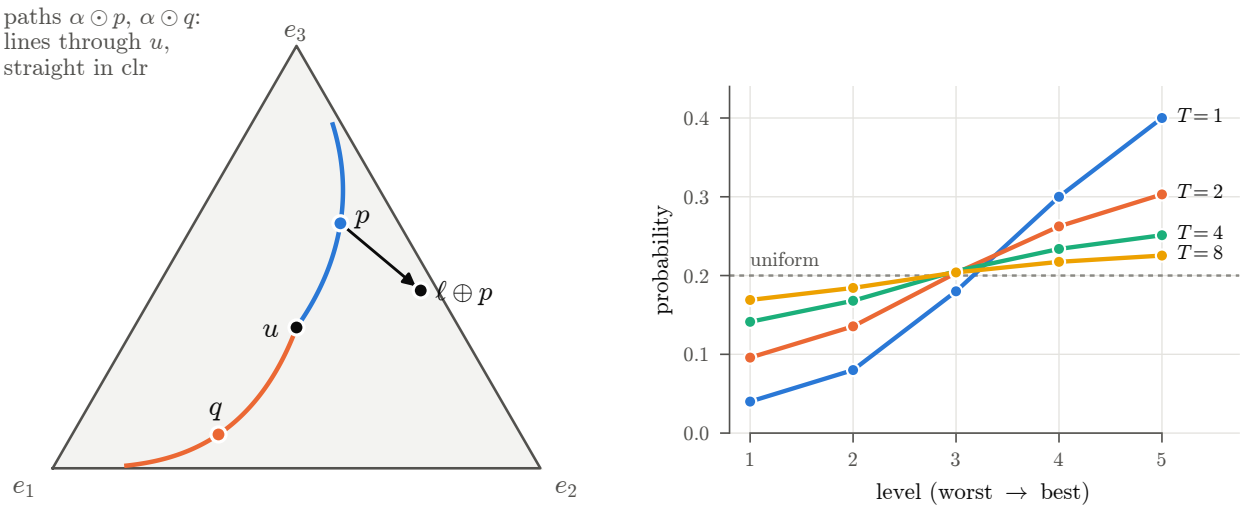
3 The three layers

3.1 Belief layer: Aitchison geometry on the simplex

Raw model output can contain exact 0s and 1s, on the boundary of the simplex where log-ratios break. The one non-linear step, $\texttt{interior}(p, \eta) = (1-\eta)p + \eta u$, moves it into the open simplex $\Delta^\circ_K = \{p : p_i > 0,\ \sum p_i = 1\}$. On Δ° define perturbation $p \oplus q = C(p \cdot q)$ (the closure C rescales to sum 1), powering $\alpha \odot p = C(p^\alpha)$, and the centred log-ratio $\mathrm{clr}(p) = \log p - \overline{\log p}$, a linear isomorphism onto $\{x : \sum x = 0\}$. Every belief operation below is linear in clr coordinates (Table 1, Figure 2).

operation	formula	meaning
Bayesian update	$\ell \oplus p$	evidence likelihood ℓ
temperature calibration	$(1/T) \odot p$	correct over-/under-confidence
log opinion pool	$\bigoplus_j (w_j \odot p_j)$	combine judges
orbit debiasing	$\frac{1}{\lvert G\rvert} \odot \bigoplus_g g^{-1} \cdot p_g$	cancel position bias

Table 1: Belief operations are linear in Aitchison geometry; only `interior` is not.



(a) The open 2-simplex. Tempering paths $\alpha \odot p$ are straight lines through the uniform point u in clr coordinates; a Bayes update $\ell \oplus p$ is a translation.

(b) Tempering an overconfident five-level belief: larger T moves it toward uniform while preserving the ranking of levels.

Figure 2: Belief layer: Aitchison geometry and calibration.

B1 (tempering is a group action). $\beta_1 \odot (\beta_2 \odot p) = (\beta_1\beta_2) \odot p$, $1 \odot p = p$; ranking is preserved for $\beta > 0$; $T \to \infty$ gives uniform. **B2.** The calibration NLL is convex in $\beta = 1/T$, so a one-dimensional search finds the global optimum. **B3.** The pool is symmetric in judges, unanimous when $\sum w = 1$, and satisfies $\text{pool}(\ell \oplus p_j) = (\sum w) \odot \ell \oplus \text{pool}(p_j)$, i.e. it is externally Bayesian exactly when $\sum w = 1$. **B4.** If the judge's bias is *multiplicative in position*, a transitive cyclic orbit Z_n removes it exactly; under reversal Z_2 (the only non-trivial re-ordering that keeps an ordinal scale ordinal) the odd part of clr(bias) cancels and the even part remains. Exactness holds for the raw answers. The pipeline first applies `interior` with $\eta > 0$, which is not multiplicative, so with the default $\eta = 0.03$ the cyclic orbit is only approximately unbiased (§6.9 quantifies the error). Bias that interacts with content is only reduced. **B5.** Hedge weights $w \propto e^{-\eta\,\text{loss}}$ are shift-invariant and antitone in loss. Derivations are in Appendix A.

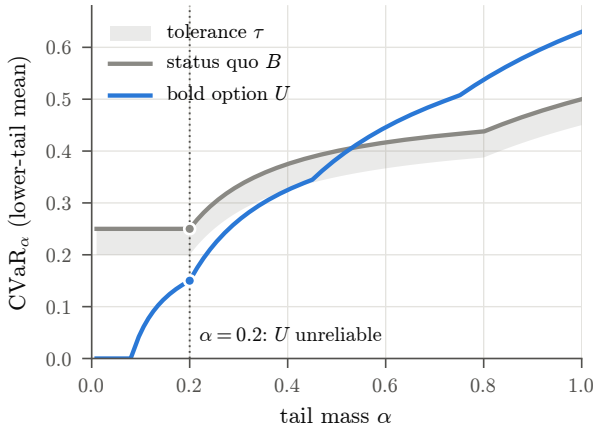
3.2 Value layer: finite lotteries

A lottery $\mu = \sum_k p_k\,\delta_{x_k}$ has sorted, distinct atoms and masses summing to 1. Binary operations combine *independent* copies, so $X + X$ is a convolution, not $2X$.

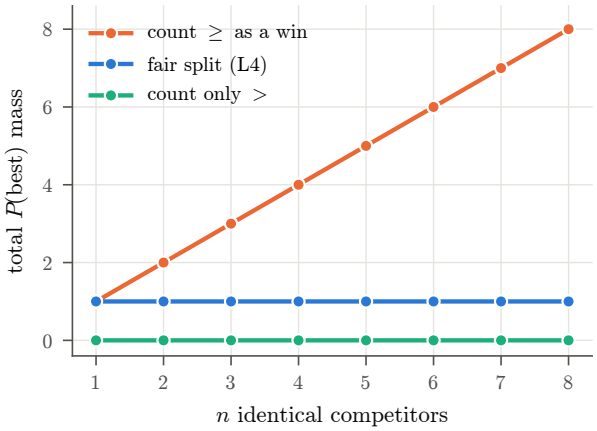
What "exact" means. All functionals are computed by enumeration, with no sampling, *for the lottery that is represented.* That lottery is a model: it inherits the judges' calibration, the chosen level utilities, and the independence assumptions. And when a convolution would exceed 4096 atoms it is compacted onto a grid, keeping mass and mean exactly but moving quantiles and CVaR by up to one grid step. Exactness is therefore a property of the arithmetic, not of the beliefs.

Definitions. For a lottery X with CDF F and quantile function F^{-1}, the lower-tail mean at level $\alpha \in (0, 1]$ is

$$\text{CVaR}_\alpha(X) = \frac{1}{\alpha}\int_0^\alpha F^{-1}(u)\,du = \max_{t\in\mathbb{R}}\Big\{t - \tfrac{1}{\alpha}\,\mathbb{E}[(t - X)^+]\Big\}, \tag{1}$$



(a) CVaR$_\alpha$ of a bold option U and the status quo B. At $\alpha = 0.2$, U falls below the band $[\mathrm{CVaR}(B) - \tau, \mathrm{CVaR}(B)]$, so it is not reliable, although its mean is higher.



(b) Total $P(\text{best})$ mass among n identical competitors. Counting $\geq$ as a win over-counts (n); counting only $>$ under-counts (0); the fair split of L4 sums to 1.

Figure 3: Value layer: the reliability test and fair tie splitting.

which takes the part of an atom that straddles the α-quantile in proportion. Two values are treated as equal when they differ by less than a relative 10^{-9}: atoms that agree to that tolerance are merged and count as ties in $P(\text{best})$ and in the argmax. Without this, rounding errors from different computation paths made equal values unequal; writing down P8 exposed the bug.

L1. $(+, \cdot)$ are commutative and associative; expectation is linear; variances add for independent sums. **L2.** Lower-tail CVaR$_\alpha$ is FSD-monotone, translation-equivariant, positively homogeneous, superadditive, bounded by the mean, monotone in α, with $\mathrm{CVaR}_1 = \mathbb{E}$. **L3.** Compaction preserves mass and mean and moves CVaR by at most one grid step; a lottery built through m compacting operations is within the sum of their grid steps.

L4 (fair-tie $P(\text{best})$). For independent competitors V_j, where competitor j is removed (value $-\infty$) with probability r_j,

$$P(\text{best}_i) = \sum_x p_i(x) \int_0^1 \prod_{j \neq i} (F_j(x^-) + p_j(x)\, t)\, dt, \qquad \int_0^1 t^k\, dt = \frac{1}{k+1}, \tag{2}$$

which averages $1/(1 + \#\text{ties})$ over the number of ties. It sums to 1, is permutation-equivariant and splits equally between identical competitors (Figure 3b). **L5.** $\mathbb{E}[\max_j U_j] = \sum_x x\, \Delta \prod_j F_j(x)$ and $\mathrm{EVPI} = \mathbb{E}[\max_j U_j] - \max_j \mathbb{E}[U_j] \geq 0$. **L6.** $\mathrm{EI}(V; B) = \mathbb{E}[\max(V - B, 0)]$ is monotone in the convex order (a mean-preserving spread never lowers it) and is at least the mean gain (Figure 5a).

3.3 Policy layer: gates and selection

For option i with dimension beliefs $p_{i,d}$, level utilities v_d, weights w_d, soft penalties λ and hard-violation probabilities $q_{i,c}$:

$$V_i = m + \Big(\sum_d \frac{w_d}{\sum w} X_{i,d} - m\Big) \prod_{\text{soft } c} M_{i,c}, \qquad M_{i,c} = \begin{cases} 1 - \lambda & \text{w.p. } q_{i,c} \\ 1 & \text{else} \end{cases}$$

$$f_i = 1 - \prod_{\text{hard } c} (1 - q_{i,c}), \qquad U_i = \begin{cases} V_i & \text{w.p. } 1 - f_i \\ B & \text{w.p. } f_i \end{cases}$$

where $m = \sum_d \frac{w_d}{\sum w} \min v_d$ is the utility floor and B is the status quo (the baseline, or doing nothing). Soft penalties shrink utility toward the floor rather than toward zero; in v0.1 they multiplied V itself, so a penalty *raised* utility whenever level values were negative and decisions depended on where zero sat (P8 below). The second line is the *perfect-check* model; §4.2 generalises it. A hard check is **clear** ($q \le \delta$), **verify** ($\delta < q < \rho$) or **reject** ($q \ge \rho$); defaults are $\delta = 0.15$, $\rho = 0.7$.

$$\text{reliable}(i) \iff \text{status} \neq \text{reject} \ \wedge \ \text{CVaR}_\alpha(U_i) \ge \text{CVaR}_\alpha(B) - \tau \tag{3}$$

$$\text{commit} = \arg\max_{\text{reliable}} \mathbb{E}[U_i] \qquad \text{ties} \to \text{status quo, then id} \tag{4}$$

$$\text{innovation} = \arg\max_{\text{reliable},\ \text{novelty}\ge\nu,\ \neq\text{commit}} \text{EI}_i \tag{5}$$

$$\text{portfolio} = P(\text{best}) \text{ over } \{B\} \cup \{\text{best reliable per niche}\} \tag{6}$$

$$\text{escalate} \iff \text{EVPI} > \kappa. \tag{7}$$

Here $\text{EI}_i = \mathbb{E}[(U_i - B)^+]$ *with U_i and B coupled*: when option i falls back, its outcome is the same draw of B, so that branch contributes zero. In closed form, with perfect checks $\text{EI}_i = (1 - f_i)\,\mathbb{E}[(V_i - B)^+]$ for independent V_i, B (the general form is in §4.2). Reading the formula with an independent copy of B in the fallback branch would credit an option for the status quo's own variance. The reliability test allows $\text{CVaR}_\alpha(U_i)$ to sit up to τ *below* the status quo's: it is a tolerance on tail loss, not a pointwise guarantee.

Properties. Each has a property-based test and a derivation in Appendix A; the derivations fixed the statements of P2, P4, P6 and P7, which earlier versions over-claimed.

P1 The baseline is reliable, so a commit always exists; $P(\text{best})$ and the portfolio each sum to 1; $\text{EVPI} \ge 0$.

P2 Permuting options permutes the outputs. Renaming them does too, except that an exact tie in $\mathbb{E}[U]$ between two non-baseline options goes to the smaller id.

P3 Scaling all weights by $c > 0$ changes nothing.

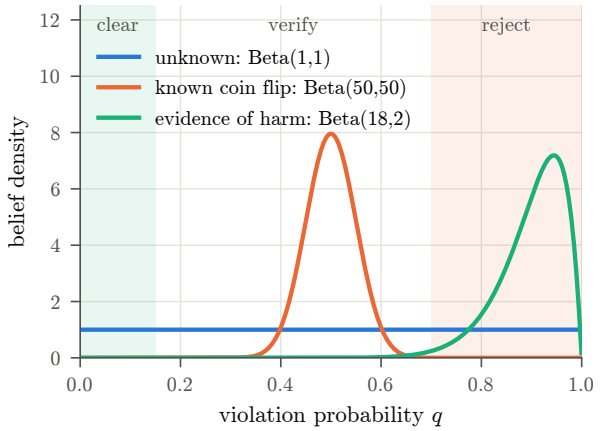
P4 Adding a rejected option, or one whose $\mathbb{E}[U]$ is strictly below the commit's, leaves the commit unchanged; exact ties go to the status quo, then to the smaller id.

P5 Novelty only gates the innovation slot.

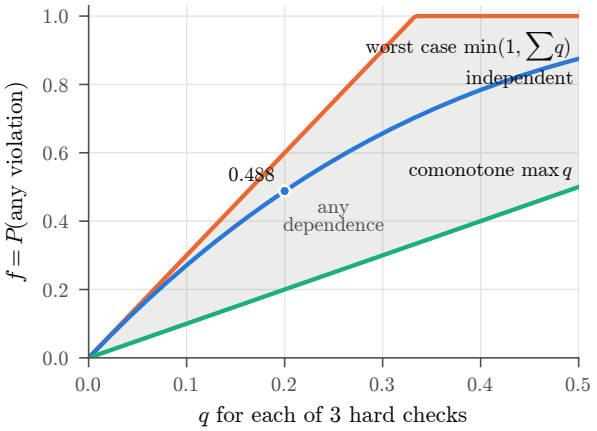
P6 Raising a hard-violation probability never improves an option's status and never lowers f. With perfect checks $\mathbb{E}[U]$ moves toward $\mathbb{E}[B]$; with imperfect checks and $H \le \min B$, EI never rises. (With imperfect checks $\mathbb{E}[U]$ need *not* move toward $\mathbb{E}[B]$: the missed-harm branch can pull it below.)

P7 With perfect checks, if $V_i \succeq_{\text{FSD}} V_j$ and $f_i \le f_j$ then $P(\text{best}_i) \ge P(\text{best}_j)$, and $\mathbb{E}[U_i] \ge \mathbb{E}[U_j]$ provided $\mathbb{E}[V_j] \ge \mathbb{E}[B]$. Without that proviso the second claim is false: two options with the same V below a better status quo reverse, because the riskier one falls back more often.

P8 Mapping every level value, H and the null value by $x \mapsto ax + b$ ($a > 0$), and τ, κ, k by $x \mapsto ax$, maps every $\mathbb{E}$, CVaR and EI the same way and leaves every gate state, $P(\text{best})$, commit, innovation and escalation unchanged.



(a) Ignorance and a known 50% harm rate share $\bar{q} = 0.5$, and so the same value of a one-shot check (Proposition 1). They differ in what a check teaches about similar options.



(b) Three hard checks, each with violation probability q. Independence gives $1 - (1 - q)^3$ (0.488 at $q = 0.2$); without assumptions f lies anywhere in the shaded band.

Figure 4: The two assumptions behind the gate: what q means, and how checks combine.

4 Uncertainty, verification and dependence

4.1 Unknown versus risky

A single number q cannot carry the whole distinction the slogan suggests. Write the belief about an option's violation probability as a distribution $\pi(q)$, for example Beta(a, b) with mean $\bar{q} = a/(a + b)$ and concentration $n = a + b$. Figure 4a shows complete ignorance Beta$(1, 1)$ and a well-measured coin flip Beta$(50, 50)$, which share $\bar{q} = 0.5$, and evidence of harm Beta$(18, 2)$. It is tempting to say that only the first is a reason to verify. For a single decision that is false.

Proposition 1 (The value of a one-shot check depends on π only through $\bar{q}$)**.** *Let a check observe whether* this *option violates, with sensitivity s, specificity c and cost k, and let a missed violation be worth H. Acting on the option after the check rather than blindly changes expected utility by*

$$\Delta_{check} = \bar{q}\, s\, (\mathbb{E}[B] - H) - (1 - \bar{q})(1 - c)(\mathbb{E}[V] - \mathbb{E}[B]) - k,$$

and verifying beats rejecting (keeping B) if and only if $(1-\bar{q})\, c\, (\mathbb{E}[V]-\mathbb{E}[B]) > \bar{q}\, (1-s)(\mathbb{E}[B]-H)+k$. *Both depend on π only through $\bar{q} = \mathbb{E}_\pi[q]$.*

The check observes the event, and $P(\text{violation}) = \mathbb{E}_\pi[q]$ by the tower rule; every branch probability is affine in it. So ignorance and a known coin flip warrant the same gate state, and `cdm` is right to act on $\bar{q}$ alone *within one decision*. The shape of π matters for *learning*. When options share a mechanism, one checked outcome updates the belief about the next one: under Beta$(1, 1)$ a single observation moves the mean from 0.5 to 0.67 or 0.33, while under Beta$(50, 50)$ it moves by 0.005. Verification under ignorance is therefore also an experiment, and its full value includes this information for future decisions. Capturing that needs judges that report uncertainty about q, and a sequential policy that keeps beliefs per mechanism. Both are outside v0.2.0.

There is an empirical point too. With a real judge, "unknown" options rarely sit at $q \approx 0.5$: on the ambiguous options of our Jev benchmark (§6.6) the median q is about 0.3, and Jev's q varies with phrasing even where the true rate does not.

4.2 Imperfect verification and residual risk

The perfect-check model assumes a check catches every violation, never raises a false alarm, costs nothing and resolves before any exposure. We drop the first three assumptions. Let a check have sensitivity $s = P(\text{caught} \mid \text{violation})$, specificity $c = P(\text{pass} \mid \text{no violation})$ and cost k, and let H be the utility of an undetected violation. Then

$$U_i = \begin{cases} V_i - k & \text{w.p. } (1-f_i)\,c \quad \text{(passes, no violation)} \\ B - k & \text{w.p. } f_i\,s + (1-f_i)(1-c) \quad \text{(caught, or false alarm)} \\ H - k & \text{w.p. } f_i\,(1-s) \quad \text{(missed)} \end{cases} \tag{8}$$

with k charged to options in the verify state. The defaults $s = c = 1$, $k = 0$ recover the perfect-check model exactly. As in the perfect-check model, the fallback applies to every hard constraint, including those in the clear state (which are assumed monitored at no explicit cost); a constraint that is not actually monitored should be modelled with $s = 0$. The policy layer needs only U_i, the race entry for $P(\text{best})$ (the fallback branch is removed from the race, since it *is* the status-quo action) and

$$\mathrm{EI}_i = (1-f_i)c\,\mathbb{E}\big[(V_i - k - B)^+\big] + f_i(1-s)\,\mathbb{E}\big[(H - k - B)^+\big],$$

with V_i and B independent inside each expectation; the fallback branch contributes nothing.

Proposition 2 (What the fallback does and does not bound)**.** *Under* (8)*, for any threshold* x*,*

$$P(U_i < x) \;=\; (1-f_i)c\,P(V_i < x+k) + \big(f_i s + (1-f_i)(1-c)\big)P(B < x+k) + f_i(1-s)\,\mathbf{1}[H < x+k].$$

In particular, with $k = 0$ *and* $x = \min B$*:* $P(U_i < \min B) = (1-f_i)c\,P(V_i < \min B) + f_i(1-s)\,\mathbf{1}[H < \min B]$*. The fallback removes the failure branch from the downside only when* $s = 1$*; the residual risk is* $f_i(1-s)$*; and the passing branch always carries* V_i*'s own lower tail, which the policy controls only through* $\mathrm{CVaR}_\alpha(U_i) \geq \mathrm{CVaR}_\alpha(B) - \tau$*.*

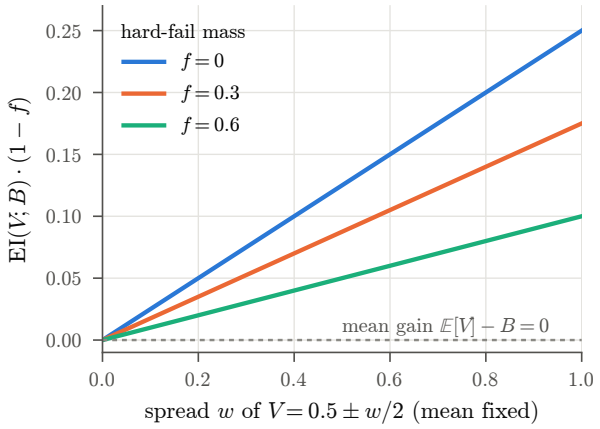
So the earlier informal claim "verify bounds the downside by construction" is true in a narrow sense: with a perfect check, the *failure* branch lands exactly on B. It is not a guarantee that realised outcomes are at least as good as the status quo. Figure 5b shows the utility distribution of a bold option as sensitivity falls. Three questions remain outside the model and must be answered per application: whether exposure can occur *before* the check resolves, whether the fallback is itself available and reliable, and how s, c and H are estimated.

4.3 Dependence between checks and between judges

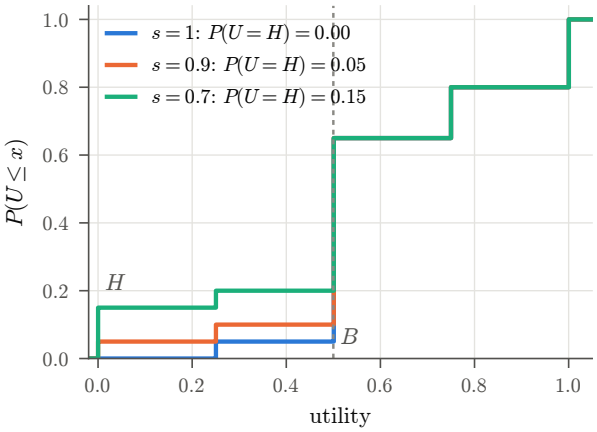
The failure mass $f_i = 1 - \prod_c (1 - q_{i,c})$ assumes independent checks. For three checks at $q = 0.2$ that gives 0.488; but without any assumption the Fréchet bounds give

$$\max_c q_{i,c} \;\leq\; f_i \;\leq\; \min\Big(1, \sum_c q_{i,c}\Big), \tag{9}$$

i.e. anywhere in $[0.2, 0.6]$ (Figure 4b). Using the upper bound is *not* a worst case for the decision: with B better than V, a larger f raises $\mathbb{E}[U]$, and a larger f can also raise CVaR (§4.4). `cdm` instead offers `hard_dependence="robust"`: since every branch weight in (8) is affine in f, $\mathbb{E}[U_i]$ is affine in f, so its minimum over all dependence structures is attained at one of the two ends, and the policy evaluates both and keeps the worse. This is a maximin rule for expected utility. CVaR is then computed at that f; it is not separately minimised (as a function of the mixing weight it is convex,



(a) Expected improvement over a fixed $B = 0.5$ for a mean-preserving spread $V = 0.5 \pm w/2$. EI rises linearly with spread and is scaled down by the hard-fail mass f.



(b) CDF of U for a bold option with $f = 0.5$ and $B = 0.5$. With a perfect check ($s = 1$) no mass falls to $H = 0$; at $s = 0.9$ and 0.7, mass $f(1 - s)$ does. Curves coincide above B.

Figure 5: Policy layer: innovation rewards spread, and the fallback is only as good as the check.

so its minimum can be interior). Gate states depend only on the marginal $q_{i,c}$ and do not change. Positive dependence (one root cause behind several violations) moves f_i *down* toward $\max_c q_{i,c}$, so independence is not automatically conservative either. A single sensitivity s is applied to the union of all hard checks; per-check sensitivities would need the joint law of the violations.

Dependence between judges is harder. With $\sum w = 1$ the log pool is a weighted geometric mean, so pooling m copies of the same judge returns that judge: the pool does not over-count agreement. But it also cannot detect that two different judges share a blind spot. If both rate a whole style of idea too highly, the pooled belief inherits the error with full confidence. We do not model this; the mitigation is to pool judges that differ in training data and architecture, and to measure their error correlation on held-out outcomes before trusting the pool.

4.4 The innovation slot and the incentive toward spread

The innovation candidate maximises expected improvement over B, which by L6 rewards spread. That is intended, since a novel idea's value is often in its upper tail, but it is also an incentive to favour volatility, so the reliability gate is what keeps it in check. Figure 6 sweeps a two-option problem: a *narrow* option and a *wide* one with the same mean, both novel, against a status quo with a slightly higher mean (so the status quo stays the commit).

Three observations follow. First, spread wins only inside a bounded region: past a spread set by τ, the wide option fails the reliability test and the slot goes to the narrow one, so the policy does not reward volatility without limit. Second, the perfect-check model contains a perverse incentive: raising q turns more of the wide option's lower tail into the status quo, which raises its CVaR, so a *riskier* volatile option can look *more* reliable (middle panel). Third, the imperfect-check model of §4.2 removes it: with $s = 0.9$ each unit of q also adds mass at H, and the region where the wide option wins no longer grows with q (right panel). We therefore recommend the check model whenever q is material, even with a rough estimate of s.

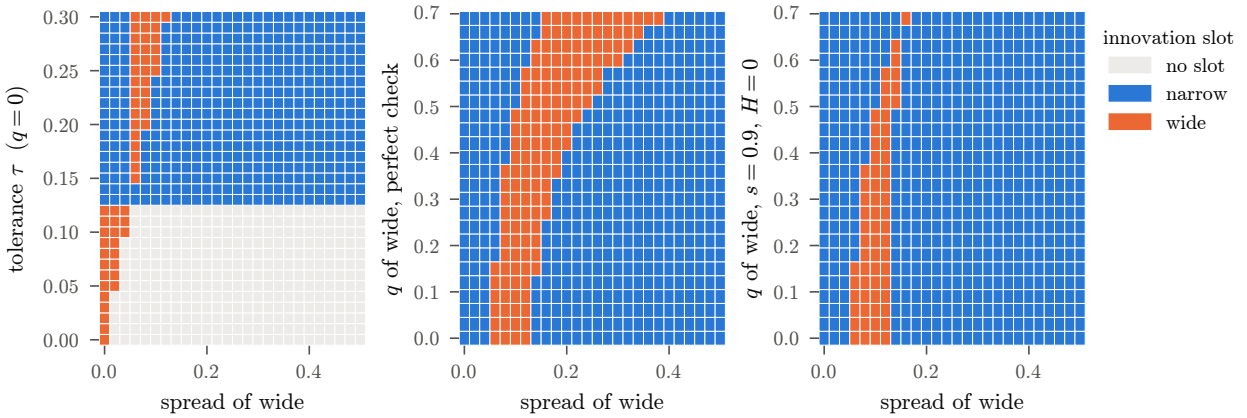


Figure 6: Which option takes the innovation slot (narrow and wide have equal means; both are novel). Left: as spread grows, the wide option's CVaR falls and it is excluded unless τ is large; at small τ nothing qualifies. Middle ($\tau = 0.3$, perfect check): a *higher* violation probability lets *wider* options pass, because failures land on B and lift the tail. Right (same, with $s = 0.9$, $H = 0$): pricing missed violations removes that artifact.

5 Creative engineering: novelty in the working mechanism

The policy of §3.3 reserves an innovation slot for options judged novel. In v0.2 that judgment was one ordinal Score ("how different is the approach from every precedent?"). Two problems follow. The Score responds to how an option is described as well as to what it does (§6.7 measures this on Jev), and it names no mechanism, so a check of one option teaches nothing about another. Engineering design has a long-standing answer to both: describe a solution by the functions it must perform and the working principle that performs each one [20, 13]. Novelty and feasibility are then properties of principles, which are shared across options and across decisions. This section adds that description as a fourth layer, `mechanism`, between the judges and the policy. Generation is still left to people or a generative model, who write the catalogue of principles; the judges classify options into it.

5.1 Functions, working principles and the precedent record

Definition 1 (Morphology)**.** A morphology is a list of functions f with weights $w_f \geq 0$ ($\sum_f w_f > 0$), each with a catalogue $\mathcal{P}_f$ of known working principles, and a record R of precedents, each mapping some functions to the principle it used. For option i a judge returns, per function, a classification $\pi_{i,f} \in \Delta(\mathcal{P}_f \cup \{\text{NEW}\})$, where NEW stands for a principle outside the catalogue, and a probability $r_{i,f}$ that the way option i performs f fails in this setting. Write $c_{f,p}$ for the number of precedents that used p for f.

Classification is a Choice question per function over the catalogue plus NEW, asked over the cyclic orbit (B4), so position bias in the catalogue order is removed as for niches; $r_{i,f}$ is a Noul. The precedents and the catalogue are what "normal" means; the previous version passed them to the judge only as free text.

5.2 Mechanism novelty and combination novelty

Definition 2 (Novelty of a mechanism)**.** With a familiarity count $m_f \geq 1$,

$$N_f(p) = 1 - \min\Big(1, \frac{c_{f,p}}{m_f}\Big), \qquad N_f(\text{NEW}) = 1,$$

$$\nu_i = \sum_f \frac{w_f}{\sum_g w_g} \sum_p \pi_{i,f}(p)\, N_f(p), \qquad \kappa_i = 1 - \sum_{r \in R^*} \prod_f \pi_{i,f}(r_f),$$

where R^* is the set of distinct precedents that record every function. ν_i is the expected novelty of the option's principles; κ_i is the probability that its *combination* of principles appears in no precedent.

The default $m_f = 1$ makes $N_f(p)$ the indicator that no precedent used p for f, so ν_i is the weighted probability that the option relies on an unused principle. Setting m_f to the number of precedents that record f gives the rarity metric of Shah, Vargas-Hernandez and Smith [17]. We do not use it by default because it depends on the size of the record: with three precedents, a principle that one of them used scores 2/3, so on our Jev data half of the options that repeat a precedent passed a 0.5 novelty gate under that metric. The two quantities correspond to two kinds of creativity distinguished by Boden [5]: ν to a new principle, κ to a new combination of known ones. Combination is also a classical source of technological novelty [3, 6], and recombination of familiar components raises the variance of outcomes [6], which is what the expected-improvement criterion of the innovation slot rewards (L6).

M1. $\nu_i, \kappa_i \in [0, 1]$; ν is affine in each $\pi_{i,f}$; moving classification mass to a principle used by fewer precedents never lowers ν; and ν depends on the option only through π, so rewording that leaves the classification unchanged leaves ν unchanged. **M2.** Adding a precedent that uses p for f never raises $N_f(p)$ and never lowers $N_g(p')$ for any other pair (it raises them under Shah's metric, whose denominator grows). In the policy, ν_i replaces the judged novelty in the eligibility test of the innovation slot; P5 still holds, since novelty enters nothing else.

5.3 Feasibility of a mechanism and the prototype gate

Let $\theta_{f,p}$ be the rate at which principle p performs f across settings, and $Z_{f,p} \sim \text{Bernoulli}(\theta_{f,p})$ whether it does in this one. The belief behind $P(Z_{f,p} = 1)$ comes, in order of precedence, from *evidence* (a $\text{Beta}(a, b)$ built from tests of p for f, shared by every option that uses p); from *settlement* ($P = 1$ if a precedent in this setting used p for f and no test contradicts it); or from *opinion* ($\text{Beta}(s_0(1 - r_{i,f}),\ s_0 r_{i,f})$ with $s_0 = 2$ pseudo-observations, for an untested or NEW principle). With independent classifications and distinct pairs per function,

$$q_i^{\text{mech}} = 1 - \prod_f \sum_p \pi_{i,f}(p)\, \bar{\theta}_{i,f,p}, \tag{10}$$

which enters the policy as one more hard constraint, "the option fails to work as described". This is the paper's principle applied to engineering: an unproven working principle has $\bar{\theta}$ near 1/2, so the option is sent to a *prototype test* with a fallback to the status quo, not vetoed; evidence of failure accumulates in $\text{Beta}(a, b)$ until the option is rejected. **M3.** $q^{\text{mech}} \in [0, 1]$, non-decreasing in each $r_{i,f}$ and non-increasing in each $\bar{\theta}$. Interactions between principles (two that work alone but not together) are not modelled; κ flags the options where that assumption matters most.

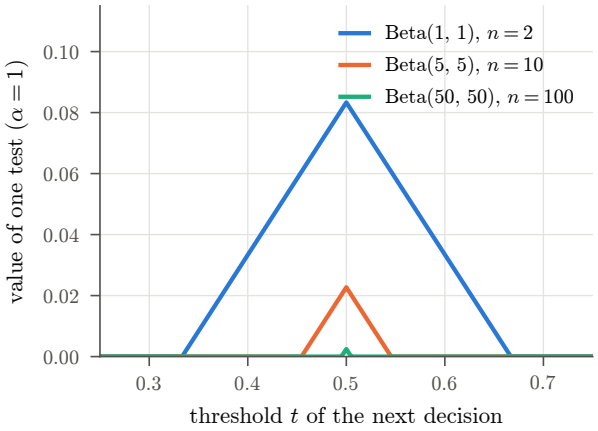


Figure 7: Proposition 3: the value of one perfect test for a later decision with threshold t, for three beliefs with mean 0.5. Each is a triangle of width $1/(n+1)$ and height $\bar{\theta}(1-\bar{\theta})/(n+1)$; ignorance ($n = 2$) is worth 34 times a known coin ($n = 100$).

5.4 Learning across decisions: what one prototype test teaches

Proposition 1 showed that within one decision only $\bar{q}$ matters, and that the shape of the belief matters for learning. Beliefs per principle make this operational. A test report Y about p with sensitivity s (to failure) and specificity c has a likelihood affine in θ, so the posterior is a mixture of $\text{Beta}(a+1, b)$ and $\text{Beta}(a, b+1)$; we store the Beta with the same mean and variance. **M4.** This update is exact (conjugate) for $s = c = 1$, leaves the belief unchanged when the test is uninformative ($s + c = 1$), and preserves the exact posterior mean, which by Proposition 1 is all that the next single decision uses. The next proposition prices what one test teaches.

Proposition 3 (Value of one test for the next decision)**.** *Let $\theta \sim \text{Beta}(a, b)$ with $n = a + b$ and $\bar{\theta} = a/n$, and let a later decision gain $\alpha(\mathbb{E}[\theta] - t)$ from acting on an option that uses p, and act if and only if that is positive. One perfect observation of Z raises the later decision's expected value by*

$$\text{VOI}(t) = \bar{\theta}\,\alpha\Big(\tfrac{a+1}{n+1} - t\Big)^+ + (1-\bar{\theta})\,\alpha\Big(\tfrac{a}{n+1} - t\Big)^+ - \alpha(\bar{\theta} - t)^+ \;\le\; \alpha\,\frac{\bar{\theta}(1-\bar{\theta})}{n+1},$$

with equality at $t = \bar{\theta}$, and $\text{VOI}(t) = 0$ unless t lies in the interval $\big(\tfrac{a}{n+1}, \tfrac{a+1}{n+1}\big)$, of width $1/(n+1)$.

Ignorance and a known coin flip, which warrant the same single decision, differ here by a factor $(n_2+1)/(n_1+1)$: for $\text{Beta}(1,1)$ against $\text{Beta}(50,50)$ the bound is $101/3 \approx 34$ times larger (Figure 7). This is the quantitative content of "verification under ignorance is also an experiment". A test of a principle that K later options share is worth at most $\sum_k \alpha_k \bar{\theta}(1-\bar{\theta})/(n+1)$. `cdm` reports, per option, $\Lambda_i = \sum_f \sum_p \pi_{i,f}(p)\,\bar{\theta}(1-\bar{\theta})/(n+1)$ over its unsettled principles, and the innovation slot can rank by $\text{EI}_i + w\Lambda_i$, where $w \ge 0$ is the utility at stake per unit of $\bar{\theta}$ in later decisions (`Policy.learning_weight`, default 0). Like novelty, $w\Lambda$ enters only the innovation slot, so P1–P8 are unchanged; under P8's affine map w scales by a.

6 Experiments

6.1 What each piece of evidence supports

We report eight kinds of evidence. Table 2 states what each supports and what it does not; only the three benchmarks compare decision rules, and only the Jev benchmark and the mechanism

experiment use a real judge.

evidence	supports	does not establish
property tests, oracles, derivations	the code implements the stated formulas, which hold under their stated hypotheses	that the formulas describe the world
exact vs. Monte Carlo	two implementations agree on the represented lottery	that the lottery is accurate
synthetic bias recovery	the pipeline removes a planted multiplicative bias	that real biases have that form
live Jev run	the pipeline works end to end on a real API	better decisions than alternatives
synthetic policy benchmark	how the gates trade regret against harm when the world matches the model	behaviour when it does not
Jev benchmark (§6.6)	how the policy performs on a real judge's judgments when the truth is known by construction	performance on real decisions and real outcomes
mechanism vs. wording (§6.7)	whether a real judge's novelty follows the working mechanism or the wording, on templated options	behaviour on real, untemplated designs
sequential benchmark (§6.8)	what learning per principle and the innovation slot do over a programme of related decisions, in a simulated world	that real principles behave like the simulated ones

Table 2: Evidence and its limits.

6.2 Implementation: property-based tests and oracles

Every numbered claim has a property-based test (Hypothesis); L4/L5 also have a brute-force oracle over joint outcomes, and `MATH.md` maps claims to tests. The gate runs `ruff`, `mypy -strict` and `pytest` with a 90% branch-coverage floor. **110 tests pass at 99% branch coverage**, plus one live test against the Jev API.

Testing and derivation found real defects, which we list because they bear on how much a passing suite proves. During development Hypothesis found an ulp overshoot in a CDF, an id tie-break that let irrelevant options flip ties, and a fixture whose "dominated" option was not quite dominated. In review, writing the derivations of Appendix A found two over-claimed properties whose tests passed only because of their fixtures (P7 for $\mathbb{E}[U]$, and P6 under imperfect checks), the utility-origin dependence of soft penalties, and that the "worst-case" dependence option was not a worst case for the decision. The test for P8 then found that rounding to 12 decimals left equal values unequal along different computation paths, which silently broke fair tie splitting; ties now use a relative tolerance (§3.2). For the mechanism layer, the first live run exposed two design errors that no test could: Shah's rarity metric made half of the repeated mechanisms "novel" because the record was small, and a Beta$(1 + c_{f,p}, 1)$ prior for principles that precedents had used put the failure risk of repeated mechanisms near 0.49. Both defaults were changed after that first analysis (§5); the results in §6.7 re-analyse the same cached answers with the new defaults, so the choice of defaults was informed by these data.

6.3 Exact vs. Monte Carlo

On the three-option problem of the functional test suite, a 400k-sample Monte Carlo of the same generative model agrees with the exact engine on every reported quantity (Table 3). This checks the arithmetic, not the beliefs it is applied to.

	$\mathbb{E}[U]$		$\text{CVaR}_{0.2}$		$P(\text{best})$	
option	exact	MC ($\pm$se)	exact	MC	exact	MC
baseline	0.5550	0.5552 ± 0.0002	0.3329	0.3327	0.524	0.525
x	0.4427	0.4432 ± 0.0003	0.1966	0.1971	0.103	0.104
y	0.5202	0.5202 ± 0.0003	0.2323	0.2320	0.373	0.371

Table 3: Exact engine vs. 400k-sample Monte Carlo (`scripts/experiments.py mc`).

6.4 A live run against Jev

With a `TYPESAFE_API_KEY` the pipeline calls the TypeSafe Jev API. The demo asks how a three-person team in Kigali should sign its first 100 merchants (6 options, 4 dimensions, 2 hard constraints, 5 niches). Table 4 reports one run of v0.2.0; an earlier run of v0.1 differed by at most 0.01 in any entry and reached the same decisions. It shows the pipeline working end to end and rejecting an option that relies on a deceptive claim; it is a single illustrative case, not evidence of decision quality.

option	$\mathbb{E}[U]$	sd	CVaR	nov	$P(\text{best})$	EI	rel.	status	niche
street	0.39	0.14	0.23	0.59	0.21	0.051	yes	verify	network
momo	0.39	0.14	0.22	0.71	0.23	0.056	yes	verify	channel
door (base)	0.36	0.12	0.21	0.21	0.34	0.000	yes	clear	direct
fake	0.36	0.12	0.21	0.74	0.00	0.000	no	reject	direct
reverse	0.35	0.13	0.19	0.70	0.11	0.029	yes	verify	demand
ads	0.29	0.11	0.17	0.11	0.10	0.025	yes	clear	paid

Table 4: Live Jev run (perfect-check model): commit = street, innovation = momo, EVPI = 0.086, so the system escalates [door, momo]. "fake" shows merchants an invented list of shops that have supposedly joined, and Jev's deception check rejects it. The baseline earns $P(\text{best})$ in every world where a verify option fails its check.

6.5 Synthetic policy benchmark

To compare decision rules rather than arithmetic, we simulate problems with a hidden ground truth and score every policy on the *same* judge beliefs, in exact expectation over check outcomes.

Simulated problems, judge and policies. Each problem has a status quo (true mean ≈ 0.5) and five candidates. A candidate's true value distribution over five levels is Dirichlet; unfamiliar candidates (50%) have more spread and upside. Each candidate is harmful with probability 0.25. For familiar candidates the judge sees harm ($q \approx 0.85$ if harmful, 0.05 if not); for unfamiliar ones it does not ($q \sim U(0.35, 0.6)$ either way). The judge's value beliefs are noisy and overconfident ($T = 1.3$). Acting on a harmful option without a check yields $H = 0$. Checks have sensitivity $s = 0.9$, specificity 0.95 and cost 0.01 unless stated. **Policies:** *raw judge* (argmax of the uncalibrated mean, no gate); *accept uncertain* (calibrated argmax over $q < \rho$, no checks); *veto uncertain* (calibrated

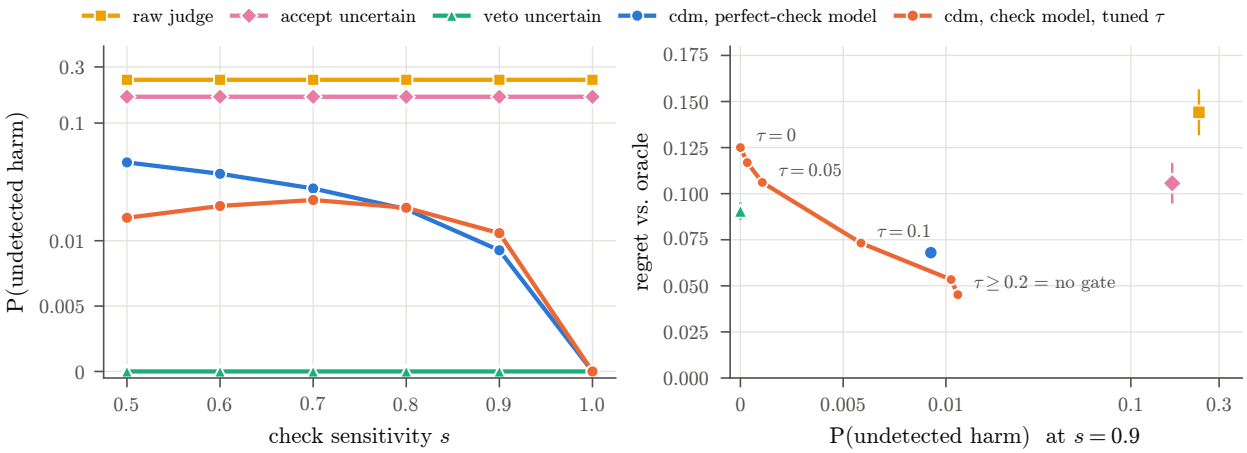


Figure 8: Synthetic benchmark, test set. Left: undetected harm as check sensitivity varies (symlog). Policies that act on uncertain options unchecked incur 17–23% harm; the perfect-check model leaks harm as s falls because it assumes its checks never miss; the check model holds it between 1% and 2%. Right: regret against harm at $s = 0.9$ as τ varies. From $\tau = 0.2$ up, the CVaR gate no longer binds and the curve stops at the no-gate point. Error bars are 95% intervals.

argmax over $q \leq \delta$, no checks); `cdm` with the perfect-check model; `cdm` with the check model (8) at the default and at a tuned τ; *expected utility + checks*, the check model with no CVaR gate ($\tau = \infty$), as an ablation; and an oracle that knows the truth. Regret is measured against the oracle.

Tuning on held-out problems. The tolerance τ is chosen on 1000 tuning problems (seed 1), from the grid $\{0, 0.025, 0.05, 0.1, 0.15, 0.2, 0.3, \infty\}$, as the smallest value with the lowest regret; it is $\tau^\star = 0.2$. Everything else is evaluated on 1500 fresh problems (seed 2). Differences against veto are paired over the same problems and reported with 95% normal intervals.

policy ($s = 0.9$, test set)	regret ($\pm$se)	undetected harm	explores	gain vs. veto [95% CI]
raw judge	0.144 ± 0.006	0.233	0.68	-0.054 $[-0.068, -0.039]$
accept uncertain	0.106 ± 0.006	0.167	0.71	-0.015 $[-0.028, -0.002]$
veto uncertain	0.091 ± 0.002	0.000	0.00	—
`cdm`, perfect-check model	0.068 ± 0.002	0.009	0.38	$+0.023$ $[0.017, 0.028]$
`cdm`, check model, $\tau = 0.05$	0.106 ± 0.002	0.001	0.04	-0.016 $[-0.019, -0.012]$
`cdm`, check model, $\tau^\star = 0.2$	0.045 ± 0.002	0.012	0.48	$+0.045$ $[0.041, 0.050]$
expected utility + checks (no gate)	0.045 ± 0.002	0.012	0.48	$+0.045$ $[0.041, 0.050]$

Table 5: Synthetic benchmark on the test set. "Explores" is how often the chosen option is an unfamiliar one (the oracle: 0.58). The tuned policy and the no-gate ablation choose identically on every test problem.

Regret and harm by policy. (1) *Checking is what makes uncertain options usable.* Accepting them unchecked explores as much as the oracle but harms in one problem in six; vetoing them is safe but forgoes the upside. Every `cdm` variant that checks uncertain options and is not starved by its tolerance beats veto. (2) *The CVaR reliability gate did not help here.* The best policy on the tuning set is the one where the gate no longer binds, and on the test set it chooses exactly as plain expected utility with checks does. Tightening τ only moves along a regret–harm frontier (Figure 8): $\tau = 0.1$ halves harm to 0.6% at a regret of 0.073; $\tau = 0.05$, the default, explores almost nothing and

is worse than vetoing. Whether the gate earns its place depends on how much a decision-maker values tail protection beyond what the check model already prices, which this benchmark does not settle. (3) *A misspecified check model leaks harm.* The perfect-check model reaches 0.9% harm at $s = 0.9$ and 4.6% at $s = 0.5$; modelling the checks keeps it between 1.2% and 2.2% at every s tested.

Robustness. Table 6 re-runs the tuned policy against veto under changes the tuning never saw. The advantage shrinks with worse or costlier checks but stays positive. A judge that is confidently wrong about 20% of familiar harmful options raises harm to about 4.7% for both policies, because neither checks options it believes are safe. The value-layer temperature barely matters: assuming $T = 1$ or $T = 2$ instead of the true 1.3 moves regret by less than 0.003, so this benchmark does not exercise the belief layer and says nothing about the value of calibration.

variation (test set)	`cdm` regret (harm)	veto regret (harm)	gain [95% CI]
none	0.045 (1.2%)	0.091 (0%)	+0.045 [0.041, 0.050]
judge confidently wrong on 20% of harmful	0.066 (4.7%)	0.116 (4.9%)	+0.051 [0.045, 0.056]
specificity 0.8	0.056 (1.1%)	0.091 (0%)	+0.035 [0.031, 0.039]
check cost 0.05	0.069 (0.8%)	0.091 (0%)	+0.022 [0.018, 0.026]
specificity 0.8 and cost 0.05	0.076 (0.6%)	0.091 (0%)	+0.014 [0.011, 0.018]
`cdm` assumes $T = 1$ / $T = 2$	0.045 / 0.048	0.091	+0.046 / +0.043

Table 6: Robustness of the tuned policy ($\tau^\star = 0.2$), paired against veto.

The simulator shares the policy's structure (binary harm, a fallback to B, a fixed H), so these results are partly a consequence of the model. Only the commit is scored: the innovation slot, portfolio and escalation are not evaluated by either benchmark.

6.6 Semi-synthetic benchmark with real Jev judgments

To test the policy on a real judge, we keep a known ground truth but let Jev do all the judging.

Templated options with known truth. Four growth problems (a meal-kit service, a climbing gym, an accounting app, a cleaning service), 12 instances each, give 48 problems. Each has a status quo ("keep doing what we do") and five candidates assembled from parts with fixed truth: a growth *mechanism*; a *value level* $v \in \{0, \dots, 4\}$ (new paying customers a month), stated in the text as a noisy two-week pilot estimate (log-normal error, $\sigma = 0.45$) or, for 30% of options, not stated at all; and a phrase about the hard constraint "mislead customers or use people's personal data without consent" that is *explicitly harmful* (true violation), *explicitly safe* (no violation) or *ambiguous*, where the deciding fact is left out and the truth is a hidden coin with $P = 0.4$. Example: "Send a weekly recipe email newsletter, using a contact list from a partner. A two-week pilot suggests about 74 new paying customers a month." Jev (`jev-latest`, October 2026) answers through `cdm`'s own pipeline: 237 unique requests, about 21 s in total, with raw answers cached so the analysis is reproducible offline. Checks and outcomes are simulated from the hidden truth as in §6.5. Nothing is tuned on these data: $\tau^\star$ is the synthetic one, and the temperatures are the audit priors.

Judgment quality. *Harm.* Jev's mean violation probability is 0.93 on explicitly harmful options, 0.15 on explicitly safe ones and 0.34 on ambiguous ones, whose true rate in the sample is 0.37 (Figure 9). AUROC is 0.9998 on explicit options and 0.92 overall; Brier score 0.104 and ECE 0.087, or 0.074 after the default Noul calibration. Two caveats matter for the gate: a fifth (21%) of the

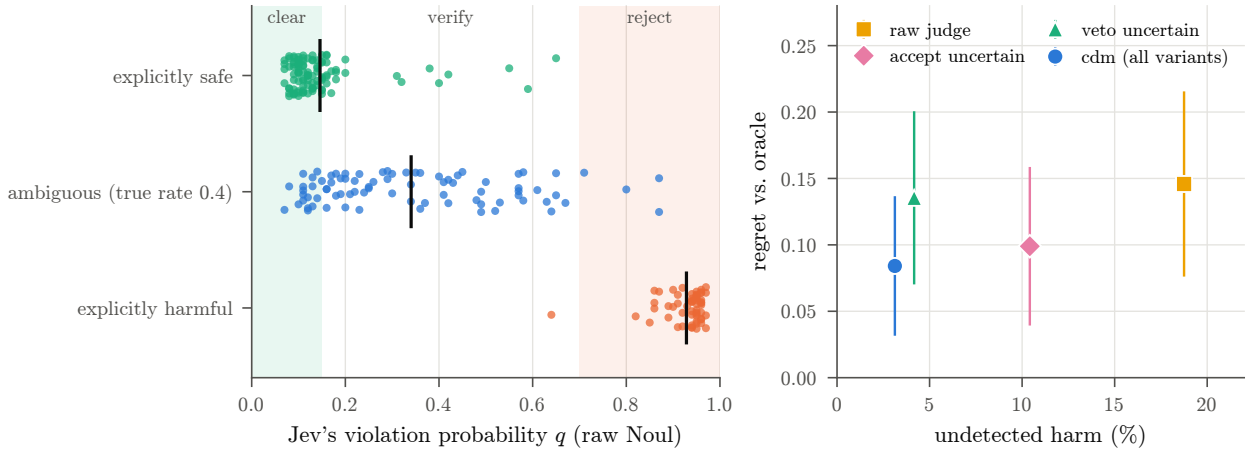


Figure 9: Jev benchmark (48 problems, 240 candidates). Left: Jev's raw violation probability by the option's true class; bars are class means. Explicit harm and explicit safety separate almost perfectly; ambiguous options spread across the verify band. Right: regret against undetected harm with 95% intervals. All four `cdm` variants choose identically.

safe options get $q > \delta = 0.15$ and so are sent to verify or vetoed for nothing; and on ambiguous options Jev's mean q ranges from 0.21 to 0.54 across phrasings although the truth rate is the same for all, so its q there reflects a prior about the phrasing, not knowledge of the case. *Value.* Jev's most likely level is right 50% of the time (63% when a pilot figure is given) and within one level 88% of the time. Held-out calibration (two folds) shows Jev to be more overconfident on this task than the audit prior assumes: per-option NLL of the true level is 1.88 raw, 1.57 with the audit's $T = 1.3$, and 1.31 with a temperature fitted on the other fold ($T \approx 2.2$–2.5). The two presentation orders of each Score differ by only 0.03 levels on average, so the reversal orbit, which doubles the Score questions, corrects almost nothing for Jev here.

policy	regret (±se)	undetected harm	picks ambiguous	gain vs. veto [95% CI]
raw judge	0.146 ± 0.036	18.8%	0.29	-0.010 $[-0.099, 0.078]$
accept uncertain	0.099 ± 0.030	10.4%	0.35	$+0.036$ $[-0.042, 0.115]$
veto uncertain	0.135 ± 0.033	4.2%	0.17	—
`cdm` (all four variants)	0.084 ± 0.027	3.1%	0.31	$+0.051$ $[0.009, 0.102]$

Table 7: Decisions on Jev's judgments (48 problems; bootstrap intervals over problems). The four `cdm` variants are the perfect-check model, the check model at $\tau = 0.05$ and $\tau^\star$, and expected utility with checks and no gate.

Decisions on Jev's judgments. `cdm` has the lowest regret and the lowest harm (Table 7). Against veto the paired gain is 0.051 with a bootstrap interval of $[0.009, 0.102]$. Veto is *not* harmless with a real judge: some harmful ambiguous options get $q \leq \delta$, and veto commits to them unchecked (4.2% harm). With these judgments the four `cdm` variants choose identically: the CVaR gate excludes only three non-rejected options across all 48 problems, none of them a winner, and the check model and τ change nothing. The benefit again comes from checking uncertain options rather than from the gate.

Caveats. The sample is small and the intervals are wide. The options are templated, and the truth is ours by construction: in particular the status quo's true level (2 of 0–4) is an assumption with no evidence in the text, and Jev judges it lower (mean utility 0.34 against 0.5). The gain depends on it: if the status quo is truly level 1 the gain over veto is 0.083 $[0.025, 0.148]$; if it is level 3 it is 0.020 $[-0.019, 0.067]$, not distinguishable from zero. Checks are simulated. This benchmark shows the policy working on a real judge's outputs; it does not show better decisions in the world.

6.7 Does novelty follow the mechanism or the wording? (Jev)

Problems, mechanisms and wordings. Four design problems, each described by two functions with a catalogue of three working principles and a record of three precedents: signing up shops for an app (find owners, collect the fee), winning meal-kit customers (first order, delivery), keeping vaccines cold on the way to remote health posts (transport, cooling) and stopping bad software releases (detect, undo). Each problem has seven mechanisms with a classification known by construction: two repeat a precedent, three use a catalogued principle that no precedent used (e.g. delivery drones; feature flags), and two use a principle outside the catalogue (e.g. sealed cool boxes handed to scheduled bus drivers; an evaporative clay-pot cooler; replaying production traffic against the new release). Each mechanism is written twice with the same mechanistic sentence: plainly, and wrapped in hype ("A radical, first-of-its-kind breakthrough that reinvents the whole approach: ... Nothing like this has been tried before."). The wording changes nothing that the true mechanism novelty ν^* depends on. Jev (`jev-latest`, October 2026) answers 56 requests through `cdm`'s pipeline, with the v0.2 novelty Score and, per function, the classification Choice and the failure Noul. Raw answers are cached; nothing is tuned.

measure	effect of hype [95% CI]	MAE vs. ν^*	Pearson r	AUROC known/novel
judged novelty (v0.2 Score)	$+0.058$ $[0.038, 0.080]$	0.122	0.90	1.00
mechanism novelty ν	$+0.005$ $[-0.006, 0.017]$	0.044	0.98	1.00
combination novelty κ	$+0.007$ $[-0.008, 0.025]$	—	—	—
mechanism risk q^{mech}	$+0.035$ $[0.018, 0.054]$	—	—	—

Table 8: Jev on 28 mechanisms × 2 wordings. Effects of hype are paired over mechanisms (bootstrap intervals); all novelty measures are on $[0, 1]$.

Classification, wording and risk. (1) *Jev identifies working principles.* The modal principle is the true one in 111 of 112 cells (99%; all catalogued cells and 15 of 16 NEW ones), and Jev puts mean probability 0.87 on the true principle, under either wording. (2) *Hype moves the ordinal novelty, not the mechanism novelty* (Table 8, Figure 10). Wrapping the same mechanism in hype raises the v0.2 novelty by 0.058 on a $[0, 1]$ scale, about a quarter of a Score level; ν, computed from the classification, moves by 0.005, an interval that includes zero, and is three times closer to the truth. (3) *The v0.2 Score was not fooled outright.* Both measures separate repeated from novel mechanisms perfectly, and neither rates a repeated mechanism novel (mean 0.08 and 0.05 against $\nu^* = 0$). The gain from ν is invariance and calibration in level, and, structurally, that it names the principle, which the learning of §5.4 needs. (4) *Hype makes Jev more sceptical that the mechanism works*: q^{mech} rises by 0.035, so hyped options are sent to prototype tests slightly more often. A gate placed exactly at a value that ν^* takes is fragile: 18 of the 20 novel mechanisms have $\nu^* = 0.5$, and at a 0.5 gate small deviations decide them (50% pass plain, 65% hyped); a gate between 0 and the smallest non-zero ν^* (e.g. 0.25) passes all novel and no repeated mechanisms under both

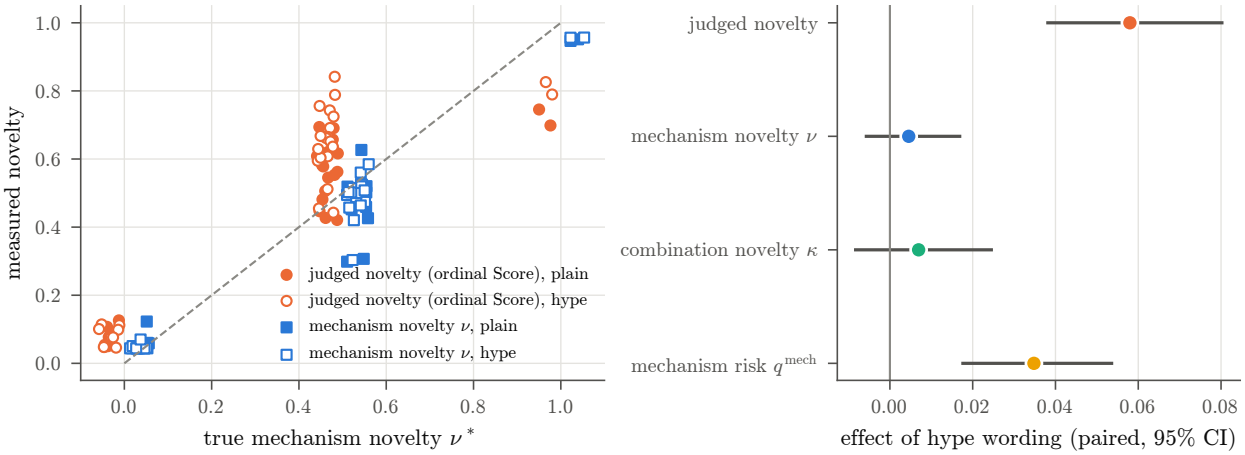


Figure 10: Left: measured against true mechanism novelty, Jev's ordinal novelty (orange) sits above the diagonal for novel mechanisms and moves up with hype wording (open markers); the mechanism novelty computed from Jev's classification (blue) stays on the diagonal under both wordings. Right: paired effect of hype wording on each quantity.

wordings. *Caveats:* the options are templated, the hype wrappers are two fixed sentences, and the classifications are easy by design because each catalogue entry is described in words close to the option's. Real options are messier, and classification accuracy must be measured per domain.

6.8 Sequential benchmark: learning per principle and the innovation slot

Simulated programme and policies. The previous benchmarks score one decision at a time and leave the innovation slot unevaluated. Here a simulated programme makes T related decisions in one domain. The design has two functions; each has two familiar principles, used by precedents and truly reliable ($\theta = 0.95$), and three untested principles with $\theta \sim \text{Beta}(1,1)$, unknown to every policy, and more upside (value contribution $U(0.3, 0.9)$ against $U(0.4, 0.6)$). Each decision offers the status quo (a familiar combination of value B) and five candidates that combine one principle per function; whether a principle works in a decision is a fresh draw $Z \sim \text{Bernoulli}(\theta)$. The simulated judge classifies principles as Jev did (probability 0.87 on the classified principle, correct 95% of the time), gives noisy, overconfident value beliefs, does not know whether untested principles work ($r \sim U(0.3, 0.6)$), and reports an ordinal novelty equal to ν^* plus 0.06 for hyped wording plus noise ($\sigma = 0.1$), the hype effect measured on Jev. Each decision commits fully to `commit` except for a share ε of the stake given to a checked pilot of the innovation option. A checked option's untested principles each produce a test report ($s = 0.9$, $c = 0.95$, cost 0.01); an option deployed unchecked reveals whether its principles worked. Payoffs are scored in expectation given the true θ against an oracle that knows θ; learning uses the sampled reports. All policies see the same worlds, candidates, judge outputs and reports, over 300 worlds per setting, and differences are paired with 95% normal intervals. Policies: *veto untested principles*; *commit only, learning* (no innovation pilot); the innovation slot with the *judged* novelty or with ν, without learning (beliefs never updated, as in v0.2); ν *with learning*; and learning with the bonus $w\Lambda$ ($w = 1, 4$). We vary $B \in \{0.5, 0.65\}$, $\varepsilon \in \{0.05, 0.2\}$ and $T \in \{24, 48\}$.

Learning, novelty measure and innovation pilot. (1) *Learning per principle pays in every setting.* Against the same policy without learning, regret falls by 0.0025 to 0.0081 (e.g. 0.0081

	$B = 0.5$, $\varepsilon = 0.05$, $T = 24$			$B = 0.65$, $\varepsilon = 0.05$, $T = 48$		
policy	regret (±se)	harm	bad / good	regret (±se)	harm	bad / good
veto untested principles	0.0542 ± 0.0019	4.1%	0.01 / 0.01	0.0083 ± 0.0006	0.6%	0.00 / 0.00
commit only, learning	0.0260 ± 0.0007	5.8%	0.12 / 0.27	0.0074 ± 0.0004	1.2%	0.02 / 0.08
judged novelty, no learning	0.0334 ± 0.0009	6.4%	0.26 / 0.22	0.0124 ± 0.0005	1.7%	0.06 / 0.05
ν, no learning	0.0331 ± 0.0009	6.4%	0.26 / 0.22	0.0124 ± 0.0005	1.7%	0.06 / 0.05
ν + learning	0.0250 ± 0.0007	5.6%	0.07 / 0.27	0.0085 ± 0.0003	1.3%	0.01 / 0.11
ν + learning + bonus ($w = 1$)	0.0246 ± 0.0007	5.5%	0.07 / 0.27	0.0089 ± 0.0004	1.3%	0.01 / 0.11

Table 9: Sequential benchmark (300 simulated worlds per setting). "Bad / good" is the share of commitments in the last eight decisions that use an untested principle with $\theta < 0.3$ / one with $\theta \geq 0.7$ and value contribution ≥ 0.6. The other settings are in the text.

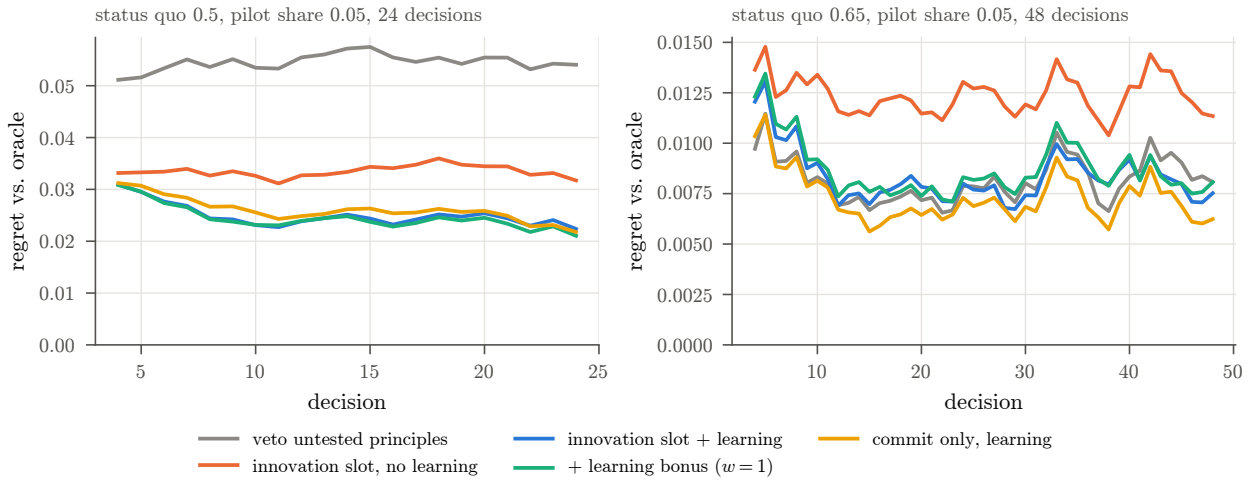


Figure 11: Regret per decision (moving average over four decisions). Left: with an ordinary status quo, learning per principle pulls regret down from the first decisions, and a small innovation pilot with learning ends slightly below committing alone. Right: with a strong status quo, vetoing untested principles is almost as good as anything, and the pilot costs more than it teaches even over 48 decisions.

$[0.0068, 0.0094]$ at $B = 0.5$, $\varepsilon = 0.05$), harm falls (from 6.4% to 5.6% there), and late commitments to unreliable principles fall from 26% to 7% ($B = 0.5$) and from 6% to 1% ($B = 0.65$). This is Proposition 3 at work: tests of untested principles move beliefs a lot, and the moved beliefs are reused by every later option that shares the principle. (2) *The novelty measure hardly matters here*: judged novelty and ν differ by at most 0.0009 in regret, because the simulated judge inherits Jev's small hype effect. What ν contributes is the named principle that learning needs. (3) *The innovation slot pays only when its pilots are small and the status quo is ordinary.* At $B = 0.5$, $\varepsilon = 0.05$, it beats committing alone by 0.0009 $[0.0001, 0.0018]$, and by 0.0014 $[0.0006, 0.0022]$ with the bonus at $w = 1$. At $\varepsilon = 0.2$ it loses 0.0034 $[0.0025, 0.0043]$; with a strong status quo it loses in every setting (-0.0011 $[-0.0015, -0.0007]$ over 48 decisions), although it raises the share of late commitments to good new principles from 0.08 to 0.11. With an ordinary status quo the commitments themselves explore, so a separate pilot mostly duplicates what they learn; with a strong one, vetoing untested principles is within 0.0008 of the best policy and halves harm. (4) *The learning bonus is not robust*: it helps slightly in the one setting where the slot pays and hurts elsewhere ($w = 4$ always does worse than $w = 1$), so its default stays 0. *Caveats:* the world shares

the model's structure (independent principles, Beta reliabilities, the same check model), the horizon is short, and payoffs are in expectation; the benchmark shows when per-principle learning and an innovation pilot can pay, not that they do on real programmes.

6.9 Bias recovery

An audit of Jev reported Choice and Score questions as overconfident ($T \approx 1.3$), Noul as underconfident ($T \approx 0.9$), and a first-option bias in Choice. `cdm` fits per-type temperature by maximum likelihood (B2 makes the fit global), uses the reversal orbit for Score questions and the cyclic orbit for Choice questions. The end-to-end test plants a multiplicative position bias in every answer and recovers the true beliefs to 10^{-9}, but it does so with smoothing off ($\eta = 0$), with $T = 1$, and with Score bias restricted to its odd part, which is exactly what the orbits remove. It is a consistency check of B4, not a validation. With the default $\eta = 0.03$, the cyclic orbit's error on a random five-level belief under a random multiplicative bias has median 0.017 and 95th percentile 0.061 (maximum absolute error per distribution, 2000 draws). On the Jev benchmark the Score temperature fitted on held-out data (≈ 2.2–2.5) differs substantially from the audit prior, so the priors should be refit per domain.

7 Discussion and limits

What is assumed. Dimensions, options and judges are treated as independent; checks are too, unless the robust dependence rule is selected. One sensitivity applies to the union of an option's hard checks. Level utilities are interval-scale policy choices, while Jev's Score levels are ordinal; the novelty measure averages ordinal levels as if they were interval. B4 is exact only for multiplicative position bias and unsmoothed answers. The gate acts on $\bar{q}$, which is sufficient for one decision but not for learning across decisions (§4.1).

What is not modelled. Exposure before a check resolves; the reliability of the fallback itself; correlated judge errors; correlations between dimensions within an option; interactions between working principles. Escalation is reported, not executed. The innovation slot is evaluated only in the simulated programme of §6.8; the portfolio and escalation are not evaluated.

Limits of the mechanism layer. The mechanism layer moves the novelty question from wording to a description that people write: the functions and the catalogue of principles. A poor catalogue (principles that are really the same, or a function split differently from how options work) produces poor novelty, and an option the judge files under NEW carries no shared evidence until someone names its principle. Principles are assumed not to interact, and classifications to be independent across functions; κ marks where the first assumption matters most. The settled rule treats a precedent in the same setting as proof that a principle works there; with precedents from other settings, record them as evidence instead.

Calibration. Temperature scaling corrects a systematic bias in confidence; it does not make a judge right, and a judge calibrated overall can be miscalibrated for a particular kind of idea, as Jev's per-phrasing q shows. The evaluation that matters is whether events assigned probability p occur with frequency p in comparable held-out cases, by domain, with sample sizes and intervals; §6.6 is a first, templated instance of that.

Tests are not proofs. Property-based tests sample a hypothesis space; they find counterexamples but cannot establish a universally quantified statement. Two properties in the previous version had passing tests and false statements. Appendix A now derives every numbered claim with its hypotheses.

Towards a real benchmark. The next step is validation on real outcomes: run `cdm` where checks are cheap and outcomes measurable (code changes behind tests, product experiments behind flags), log every judgment, check and outcome, and compare against the raw judge, calibration without the policy, and plain expected utility with checks, on regret, constraint violations, realised innovation and cost. The same log closes the loop the architecture is built for: refit temperatures and judge weights, estimate s and c per check, and decide again.

8 Conclusion

`cdm` keeps generation apart from selection and makes the selection rule explicit: calibrated beliefs in Aitchison geometry, exact arithmetic on finite lotteries, a policy that verifies uncertain options against a fallback instead of vetoing them, and a mechanism layer that measures novelty on how an option works and learns, per working principle, from every test. The principle holds under conditions we now state: the check must be informative enough to pay for itself (Proposition 1), its misses must be priced, and the tolerance for tail loss must be chosen with that pricing in view. In both benchmarks the benefit came from checking uncertain options; the CVaR reliability gate added nothing measurable, and whether it earns its place is an open question. The mechanism layer makes novelty a property of a design's working principles rather than its wording, which a real judge can supply reliably, and turns each prototype test into evidence about every option that shares the principle; in simulation that learning pays in every setting, while a separate innovation pilot pays only when it is small. The math is derived and tested; whether the policy makes better real decisions remains to be measured.

Reproducibility. Source, tests and the configuration behind every number here: https://github.com/chrisnkuno/creative-decision, tag `v0.3.0`, MIT licence. Install from source with `uv sync`; the name `cdm` on PyPI belongs to an unrelated project.

- `./scripts/check.sh`: format, lint, strict types, tests and the coverage floor;
- `uv run python scripts/experiments.py`: the benchmark and sweeps (fixed seed);
- `uv run python scripts/jev_benchmark.py collect`: the Jev benchmark's API calls (needs `TYPESAFE_API_KEY`); `analyze` reruns the analysis offline from the cached answers;
- `scripts/mechanism_jev.py collect` (needs the key) and `analyze`: the mechanism-versus-wording test; `scripts/mechanism_benchmark.py`: the sequential benchmark;
- `uv run python scripts/figures.py`: every figure, and `scripts/verify_figures.py` cross-checks every number quoted from them;
- `uv run python demo.py`: the live example (needs `TYPESAFE_API_KEY`; it falls back to a mock judge without one);
- `MATH.md`: the claim-to-test map.

A Derivations

Notation: C is closure, $\odot$ powering, $\oplus$ perturbation. All lotteries are finite.

B1. $\beta_1 \odot (\beta_2 \odot p) = C\big(C(p^{\beta_2})^{\beta_1}\big) = C(p^{\beta_1\beta_2})$ because C ignores positive scalar factors. $x \mapsto x^\beta$ is increasing for $\beta > 0$, so ranks are preserved; $p^\beta \to \mathbf{1}$ as $\beta \to 0$, so $C(p^\beta) \to u$.

B2. For labelled data (p_n, y_n), $\mathrm{NLL}(\beta) = \sum_n \big[-\beta \log p_{n,y_n} + \log \sum_k e^{\beta \log p_{n,k}}\big]$, a sum of log-sum-exp functions of linear functions of β minus a linear term, hence convex.

B3. $\mathrm{pool}(p) = C(\prod_j p_j^{w_j})$. Then $\mathrm{pool}(\ell \oplus p_j) = C(\prod_j (\ell p_j)^{w_j}) = C(\ell^{\sum w} \prod_j p_j^{w_j}) = (\sum w) \odot \ell \oplus \mathrm{pool}(p)$.

B4. Suppose that under presentation g label i is shown at position $g(i)$ and the judge reports $p_g(i) \propto b_{g(i)}\, p(i)$. The orbit average is $\propto p(i) \big(\prod_{g \in G} b_{g(i)}\big)^{1/|G|}$. If $G = Z_n$ acts transitively, each label visits each position once, so the product is $\prod_k b_k$ for every i and is removed by closure. For $G = Z_2$ (reversal $k \mapsto n-1-k$) the factor is $(b_k b_{n-1-k})^{1/2}$: in clr coordinates the odd part of $\mathrm{clr}(b)$ cancels and the even part remains.

L2. By the dual form $\mathrm{CVaR}_\alpha(X) = \min\{\mathbb{E}_Q[X] : dQ/dP \le 1/\alpha\}$ [15], CVaR is a minimum of linear functionals, hence superadditive and positively homogeneous; translation-equivariance and FSD-monotonicity follow from the quantile form, which also shows it is the mean of the worst α of outcomes, increasing in α and equal to $\mathbb{E}$ at $\alpha = 1$.

L3. Compaction moves each atom to one of the two neighbouring grid points with the probabilities that keep its mean, so there is a coupling with $|X - X'| \le h$ (the grid step); CVaR is 1-Lipschitz under such couplings, so it moves by at most h. Sums of independent lotteries and multiplication by factors in $[0, 1]$ do not enlarge a coupling's sup-distance, so errors from m compactions add up to at most $\sum_k h_k$.

L4. Fix i at value x. Each other competitor is independently below x (or removed) with probability $F_j(x^-)$ and tied with probability $p_j(x)$. Given K ties, a fair split gives i the share $1/(1+K) = \int_0^1 t^K\, dt$. Taking expectations over the independent tie indicators and exchanging sum and integral gives $\int_0^1 \prod_{j \ne i} (F_j(x^-) + p_j(x) t)\, dt$. Summing over i recovers $P(\text{some competitor attains the max}) = 1$ because some competitor never fails.

L5. $P(\max_j U_j \le x) = \prod_j F_j(x)$ by independence, so $P(\max = x) = \prod_j F_j(x) - \prod_j F_j(x^-)$. Since $U_j \le \max_k U_k$ pointwise, $\mathbb{E}[U_j] \le \mathbb{E}[\max_k U_k]$ for every j, so EVPI ≥ 0, with equality when all lotteries are point masses and one of them is the maximum.

L6. $x \mapsto (x-b)^+$ is convex, so $V' \succeq_{cx} V$ implies $\mathbb{E}[(V'-b)^+] \ge \mathbb{E}[(V-b)^+]$ for each b; integrate over the independent B. And $(x)^+ \ge x$ gives $\mathrm{EI} \ge \mathbb{E}[V] - \mathbb{E}[B]$.

P1. The baseline is always clear and $U_B = B$, so $\mathrm{CVaR}_\alpha(U_B) \ge \mathrm{CVaR}_\alpha(B) - \tau$ for every $\tau \ge 0$: the reliable set is non-empty.

P2. Every per-option quantity is computed from that option's judgment and B alone; $P(\text{best})$, EVPI and the portfolio are symmetric functions of the competitors (L4, L5). The only place names enter is the argmax tie-break, which uses ids after the status-quo rule, so a renaming can change the outcome only through an exact tie between two non-baseline options.

P3. V_i depends on weights only through $w_d / \sum w$, and so does m.

P4. Adding an option does not change B, hence not the reliability threshold or the reliability of existing options. A rejected option is not reliable. A reliable option with $\mathbb{E}[U]$ strictly below the commit's cannot be the argmax. (Strict FSD-dominance implies a strictly smaller mean.) If the added option ties the commit exactly, the status quo wins the tie if it is involved, and otherwise the smaller id does, so the commit can change; the property test uses a fresh id that sorts last.

P5. Novelty enters only the eligibility set of the innovation argmax; no lottery, gate, $P(\text{best})$, commit, portfolio or EVPI depends on it.

P6. Each gate threshold compares q with fixed constants, and $f = 1 - \prod(1 - q_c)$ is increasing in each q_c. With perfect checks $\mathbb{E}[U] = (1-f)\mathbb{E}[V] + f\,\mathbb{E}[B]$, a convex combination whose weight on $\mathbb{E}[B]$ grows with f. With imperfect checks, $\mathrm{EI} = (1-f)c\,\mathbb{E}[(V-k-B)^+] + f(1-s)\,\mathbb{E}[(H-k-B)^+]$ and the second expectation is 0 when $H \le \min B$, so EI is non-increasing in f. The counter-claim for $\mathbb{E}[U]$: $\partial\mathbb{E}[U]/\partial f = -c\,\mathbb{E}[V] + (s+c-1)\,\mathbb{E}[B] + (1-s)H$, which can be negative even when $\mathbb{E}[V] < \mathbb{E}[B]$ (e.g. $s = 0.5$, $c = 1$, H low).

P7. With perfect checks, $\mathbb{E}[U_i] - \mathbb{E}[U_j] = (1-f_i)(\mathbb{E}V_i - \mathbb{E}V_j) + (f_j - f_i)(\mathbb{E}V_j - \mathbb{E}B)$; the first term is ≥ 0 by dominance and the second is ≥ 0 when $\mathbb{E}V_j \ge \mathbb{E}B$. For $P(\text{best})$, let R_i be option i's race value (V_i w.p. $1-f_i$, $-\infty$ otherwise); $V_i \succeq V_j$ and $f_i \le f_j$ give $R_i \succeq_{\mathrm{FSD}} R_j$. Fix the other competitors and let $h(a,b)$ be the expected fair share of a competitor at value a facing one at b. Then h is non-decreasing in a and non-increasing in b, so $\phi(a,b) = h(a,b) - h(b,a)$ is non-decreasing in a and antisymmetric. Hence, with R'_j an independent copy of R_j, $P(\text{best}_i) - P(\text{best}_j) = \mathbb{E}[\phi(R_i, R_j)] \ge \mathbb{E}[\phi(R'_j, R_j)] = 0$.

P8. Under $x \mapsto ax + b$ every level value, hence every $X_{i,d}$, and the floor m map affinely; because weights are normalised the weighted sum maps the same way, and the soft factors act on $V - m$, which only scales. B, H and the null value map the same way, so every branch of (8) does, with k scaled. $\mathbb{E}$ and CVaR are translation-equivariant and positively homogeneous; EI, EVPI and the CVaR tolerance scale by a, as do τ, κ, k; and $P(\text{best})$ depends only on the order of values and on ties, which an increasing affine map preserves (up to the 10^{-9} relative tie tolerance, which is not itself affine-invariant).

Proposition 1. Blind action yields $(1-\bar{q})\mathbb{E}[V] + \bar{q}H$; the checked option yields (8) with $f = \bar{q}$; subtracting gives Δ_{check}, and subtracting $\mathbb{E}[B]$ from (8) gives the verify-vs-reject condition. The violation indicator Z satisfies $P(Z=1) = \mathbb{E}_\pi[q]$, and each branch probability is affine in $P(Z=1)$, so higher moments of π cannot enter.

Robust dependence. The set of attainable f for given marginals is the Fréchet interval (V3 below). Each branch weight of (8) is affine in f, so $\mathbb{E}[U_i]$ is affine in f and its minimum over the interval is at an end.

V1 / Proposition 2. Equation (8) is a three-branch mixture with weights $(1-f)c$, $fs + (1-f)(1-c)$, $f(1-s)$, which sum to 1. The CDF of a mixture is the mixture of CDFs; the stated expressions follow, and the H branch contributes an atom of mass $f(1-s)$ at $H-k$.

V2. If $H \le \min B$, raising s moves mass $f\,\Delta s$ from the atom at $H-k$ to the lottery $B-k$, which first-order stochastically dominates it; $\mathbb{E}$ and CVaR_α are FSD-monotone (L2).

V3. Boole's inequality gives $P(\cup_c A_c) \le \min(1, \sum_c P(A_c))$, and $P(\cup_c A_c) \ge \max_c P(A_c)$ by monotonicity; both bounds are attained by some joint distribution, so they are the best possible without further assumptions.

M1. $N_f(p) \in [0,1]$, so ν_i is a convex combination of numbers in $[0,1]$; it is affine in each $\pi_{i,f}$ because N_f does not depend on π. $N_f(p)$ is non-increasing in $c_{f,p}$, so moving mass from p to p' with $c_{f,p'} \le c_{f,p}$ cannot lower ν_i. For κ: the events "the combination equals r" for distinct $r \in R^*$ are disjoint and, by independence across functions, have probabilities $\prod_f \pi_{i,f}(r_f)$; their sum is at most 1.

M2. The new precedent raises $c_{f,p}$ by one and no other count. With m_f fixed, $N_f(p)$ cannot rise and every other N is unchanged. With $m_f = C_f$ (Shah), C_f also rises by one, which raises $N_f(p') = 1 - c_{f,p'}/C_f$ for every $p' \ne p$.

M3. Each factor of (10) is $\sum_p \pi_{i,f}(p)\bar{\theta}_{i,f,p} \in [0,1]$ with non-negative coefficients, so the product is in $[0,1]$ and non-decreasing in each $\bar{\theta}$; $r_{i,f}$ enters only through $\bar{\theta} = 1 - r_{i,f}$ for opinion-based principles.

M4. $P(Y = \text{pass} \mid \theta) = \theta c + (1-\theta)(1-s)$ and $P(Y = \text{fail} \mid \theta) = \theta(1-c) + (1-\theta)s$ are affine

in θ, so $\mathrm{Beta}(a,b)$ times either is a two-component mixture of $\mathrm{Beta}(a+1,b)$ and $\mathrm{Beta}(a,b+1)$ with weights proportional to $\bar{\theta}$ times the likelihood of the report under success and $1-\bar{\theta}$ times that under failure. For $s=c=1$ one weight vanishes (conjugacy). If $s+c=1$ the likelihood does not depend on θ and the posterior is the prior. The stored Beta has the mixture's first two moments, which exist and satisfy $\mathrm{Var} < \mu(1-\mu)$ for any non-degenerate law on $[0,1]$, so its parameters are positive.

Proposition 3. After one observation the posterior mean is $m_1 = (a+1)/(n+1)$ with probability $\bar{\theta}$ and $m_0 = a/(n+1)$ otherwise; $\mathbb{E}[M] = \bar{\theta}$ and $m_1 - m_0 = 1/(n+1)$. With $g(x) = \alpha(x-t)$, $\mathrm{VOI} = \mathbb{E}[g(M)^+] - g(\bar{\theta})^+ \geq 0$ by Jensen. If $t \notin (m_0, m_1)$, g has one sign on $\{m_0, m_1\}$, $x \mapsto g(x)^+$ is affine there, and $\mathrm{VOI} = 0$. If $m_0 < t \leq \bar{\theta}$, $\mathrm{VOI} = \bar{\theta} g(m_1) - g(\bar{\theta}) = (1-\bar{\theta})\alpha(t-m_0)$, increasing in t; if $\bar{\theta} \leq t < m_1$, $\mathrm{VOI} = \bar{\theta}\alpha(m_1 - t)$, decreasing. The maximum is at $t = \bar{\theta}$, where $\bar{\theta} - m_0 = a/n - a/(n+1) = \bar{\theta}/(n+1)$ gives $\alpha\bar{\theta}(1-\bar{\theta})/(n+1)$.

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